IMPEDANCE

PULSE TRANSIT TIME

# IM-PULSE

## EDL Final Evaluation 2026

IIT Bombay | Group MON-04

### TEAM MEMBERS


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## FINAL PROTOTYPE

*Internal PCB Assembly — STM32 + MAX30001 (x2) + ESP32*

**MON-04**

# IM-PULSE

## WHAT — The Device

A compact portable unit that simultaneously acquires ECG and dual Arterial Pulse Waveforms (APW) from bioimpedance measurements.

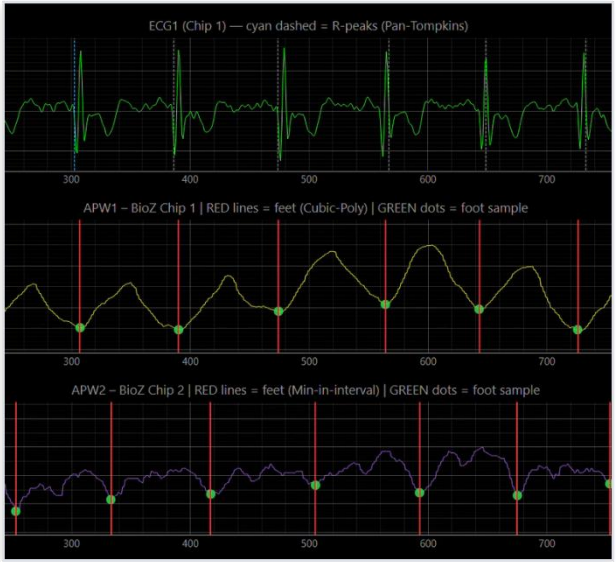
## WHO — The User

Patients who require continuous or frequent cardiac health monitoring outside clinical settings & in use cases where a compact, portable device is preferred over bulky machines.

## SOLVES — The Problem

Conventional BP monitoring relies on inflatable cuffs which can be sometimes uncomfortable for the user.

And in most cases ECG and Bio-Z readings are taken by bulky instruments.



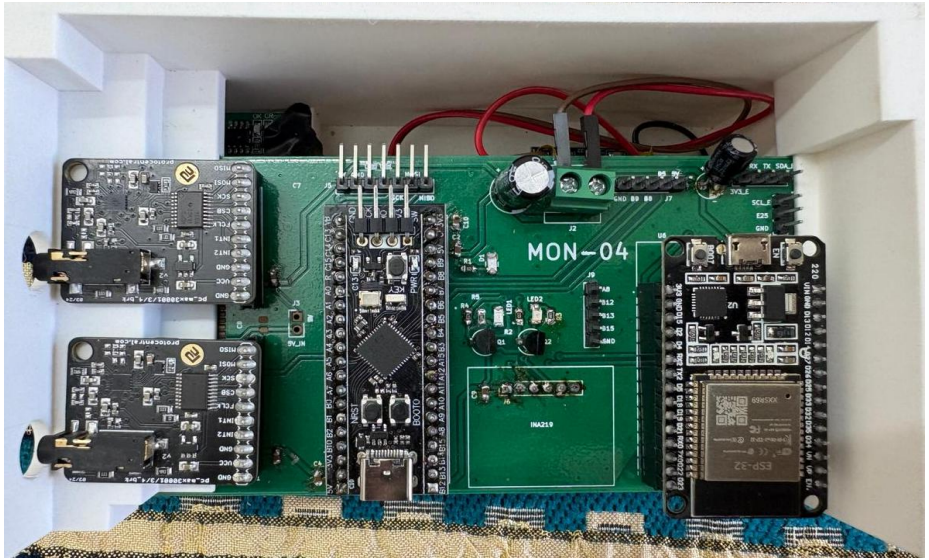
# PROJECT STATUS



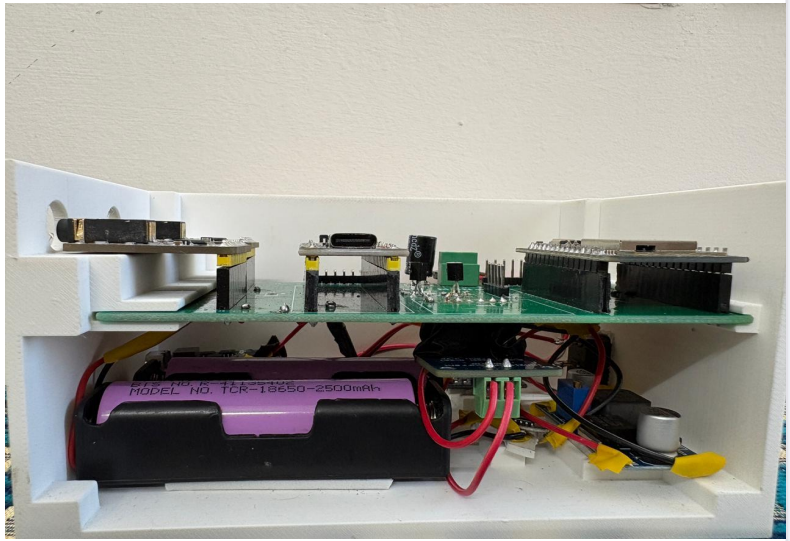
| Specification / Requirement | Status at end of the semester | Remarks / Justification |
|---|---|---|
| 4-electrodes only, two on each wrist | Incomplete | Wrist APW unviable — instead we used 6 electrodes (chest (2) + foot (2) + abdomen reference (2)) |
| Compact portable table-top unit (for bio-signal acquisition, processing & wireless) | Complete | |
| Onboard ECG R-peak & APW foot detection on STM32 | Complete | |
| Plot ECG + proximal & distal APW, verify reliability | Partial | Chest ECG & APW reliable; foot APW less regular due to lower SNR — heart rate cross-verified with clinical device |
| Obtain Pulse Wave Velocity in typical range (4–10 m/s) | Complete | |
| Derive other health parameters from PTT and cross-verify them | Partial | We focused on blood pressure as research has shown negative PTT-BP correlation; model trained on single subject (data limitation) |

# FINAL PROTOTYPE

*Fully Integrated Embedded System*




**PCB Top View — ESP32 | STM32 | 2× MAX30001 | INA219 | Custom PCB**

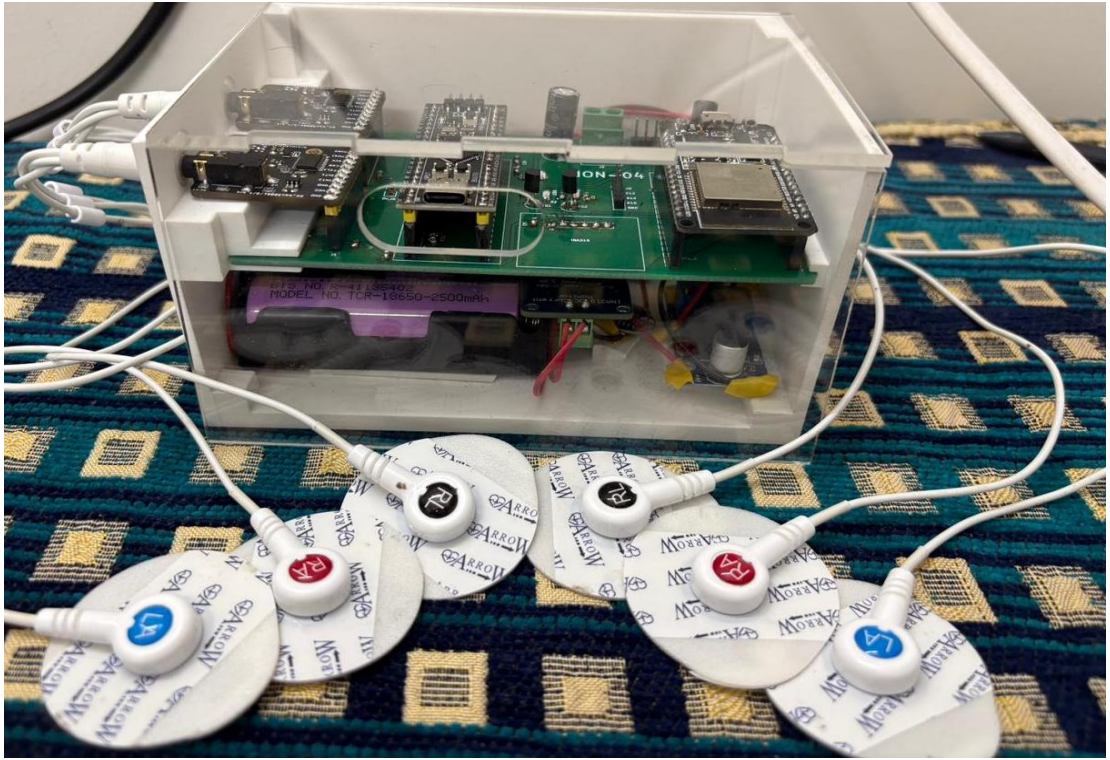



Side View — Battery Circuit | BMS Layer | 2× 18650 Cells

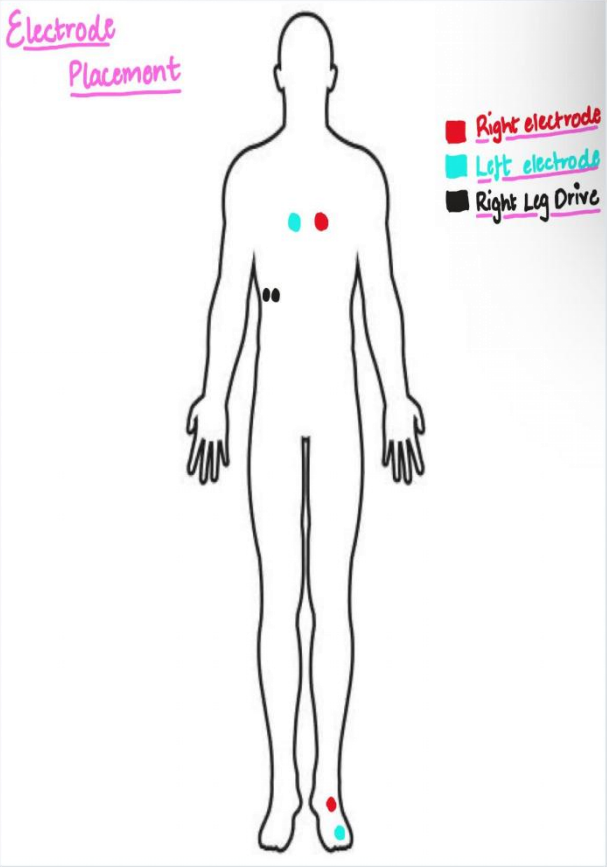
ECG + IPG Acquisition → On-board DSP (STM32) → PTT + BP Computation → Wireless Output (ESP32 → Laptop)

# FINAL PROTOTYPE

*Complete Assembled System — Device + Electrode Set*




**Assembled Device with Electrode Harness — Dual-channel IPG + ECG lead set (6 electrodes)**



# CHALLENGES & SOLUTIONS

## Dual-Purpose BioZ (Respiration + APW for PTT)

**Problem:** MAX30001 BioZ mixes respiration and blood-related IPG; respiration-focused acquisition causes DC offset and saturation, distorting APW needed for PTT.

**Solution:** Reconfigured for blood IPG (APW) by removing respiration-centric filtering, applying band-pass (~0.8–5 Hz), and syncing with ECG R-peaks to compute PTT.

## Electrode Placement & Signal Integrity

**Problem:** Accurate proximal–distal APW required precise electrode placement, and signals were distorted by environmental interference.

**Solution:** Finalized electrodes over chest (proximal) and left foot (distal) arteries; optimized placement for stable APW, and eliminated distortion by grounding nearby charged interference sources.

## PTT–BP Modeling with Custom Dataset

**Problem:** No reliable dataset exists for heart-to-foot PTT vs BP correlation, and raw measurements are noisy and unstructured.

**Solution:** Collected ~200 samples using reference BP measurements, processed and cleaned raw PTT data, and trained a model to learn PTT–BP mapping and extract patterns.